1 Questions

1. Given the curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$, determine:

   (a) The area under the curve on the given interval bounded by the x-axis.

   (b) If the area in (a) is revolved about the x-axis, determine the resulting volume.

   (c) Determine the centroid of the solid in (b) without using Pappus' theorem.

   (d) If the interval $-1 \leq x \leq 1$ is removed, determine the arc length of the curve in first quadrant, without using integration.

   (e) Based on the arc obtained in (d), if it is revolved about the y-axis, determine the centroid of the resulting surface area using Pappus' theorem.

2. Using integration, determine the arc length of the curve $y = 5x$ on the interval $0 \leq x \leq 1$.

2 Solutions

1. (a) The area under the curve can be calculated using the formula

$$A = \int_{-1}^{1} \sqrt{4 - x^2}\, dx$$

Let $x = 2\sin\theta$, then $dx = 2\cos\theta d\theta$ and the limits of integration change to $\theta = -\pi/6$ and $\theta = \pi/6$. Thus,

$$\begin{aligned}
A &= \int_{-\pi/6}^{\pi/6} \sqrt{4 - 4\sin^2\theta} \cdot 2\cos\theta\, d\theta \\
&= \int_{-\pi/6}^{\pi/6} 4\cos^2\theta\, d\theta \\
&= 4\int_{-\pi/6}^{\pi/6} \cos^2\theta\, d\theta \\
&= 4\int_{-\pi/6}^{\pi/6} \frac{1 + \cos(2\theta)}{2}\, d\theta \\
&= 2\int_{-\pi/6}^{\pi/6} (1 + \cos(2\theta))\, d\theta \\
&= 2\left[\theta + \frac{\sin(2\theta)}{2}\right]_{-\pi/6}^{\pi/6} \\
&= 2\left[\frac{\pi}{6} + \frac{\sin(\pi/3)}{2} - \left(-\frac{\pi}{6} + \frac{\sin(-\pi/3)}{2}\right)\right] \\
&= 2\left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} + \frac{\pi}{6} + \frac{\sqrt{3}}{4}\right] \\
&= 2\left[\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right] \\
&= \frac{2\pi}{3} + \sqrt{3}
\end{aligned}$$

   (b) The volume of the solid obtained by revolving the area under the curve about the x-axis can be calculated using the formula

$$\begin{aligned}
V &= \pi \int_{-1}^{1} (\sqrt{4-x^2})^2 \, dx \\
&= \pi \int_{-1}^{1} (4-x^2) \, dx \\
&= \pi \left[4x - \frac{x^3}{3}\right]_{-1}^{1} \\
&= \pi \left[(4-\frac{1}{3}) - (-4+\frac{1}{3})\right] \\
&= \pi \left[\frac{11}{3} + \frac{11}{3}\right] \\
&= \frac{22\pi}{3}
\end{aligned}$$

(c) The centroid of the solid can be calculated using the formula

$$\begin{aligned}
\bar{y} &= \frac{\frac{1}{2}\int_a^b [f(x)]^2 \, dx}{\int_a^b f(x) \, dx} \\
&= \frac{\frac{1}{2}\int_{-1}^{1} (\sqrt{4-x^2})^2 \, dx}{\int_{-1}^{1} \sqrt{4-x^2} \, dx} \\
&= \frac{\frac{1}{2}\int_{-1}^{1} (4-x^2) \, dx}{\frac{2\pi}{3} + \sqrt{3}} \\
&= \frac{\frac{1}{2} \cdot \frac{22\pi}{3}}{\frac{2\pi}{3} + \sqrt{3}} \\
&= \frac{11\pi}{3} \cdot \frac{1}{\frac{2\pi}{3} + \sqrt{3}} \\
&= \frac{11\pi}{3} \cdot \frac{3}{2\pi + 3\sqrt{3}} \\
&= \frac{11\pi}{2\pi + 3\sqrt{3}}
\end{aligned}$$

Since the solid is symmetric about the y-axis, then $\bar{x} = 0$. Thus, the centroid of the solid is at $\left(0, \frac{11\pi}{2\pi+3\sqrt{3}}\right)$.

(d) The arc length of the curve in the first quadrant is one quarter of the circumference of a circle with radius 2, which can be calculated using the formula

$$K = \frac{1}{4} \cdot 2\pi r = \frac{1}{4} \cdot 2\pi \cdot 2 = \pi$$

(e) The centroid of the surface area obtained by revolving the first-quadrant arc about the y-axis can be found using Pappus together with the surface-centroid formula. Parametrize the quarter-circle by

$$x = 2\cos\theta, \qquad y = 2\sin\theta, \qquad 0 \le \theta \le \tfrac{\pi}{2},$$

so that $ds = 2\,d\theta$ and the arc length is $L = \int_0^{\pi/2} 2\,d\theta = \pi$ (as in (d)).

The centroid of the generating arc (distance from the y-axis) is

$$\bar{x}_{\text{arc}} = \frac{1}{L}\int x \, ds = \frac{1}{\pi}\int_0^{\pi/2} (2\cos\theta)(2\,d\theta) = \frac{4}{\pi}.$$

By Pappus the surface area is

$$S = L \cdot (2\pi \bar{x}_{\text{arc}}) = \pi \cdot 2\pi \cdot \frac{4}{\pi} = 8\pi.$$

To locate the centroid of the surface (distance from the y-axis) compute

$$\bar{x}_{\text{surf}} = \frac{1}{S}\int x\,dS = \frac{1}{S}\int (2\pi x)\,x\,ds = \frac{2\pi \int x^2\,ds}{S}.$$

Here

$$\int x^2\,ds = \int_0^{\pi/2} (2\cos\theta)^2 (2\,d\theta) = 8\int_0^{\pi/2} \cos^2\theta\,d\theta = 8 \cdot \frac{\pi}{4} = 2\pi.$$

Hence

$$\bar{x}_{\text{surf}} = \frac{2\pi \cdot 2\pi}{8\pi} = \frac{\pi}{2}.$$

Therefore the centroid of the resulting surface is located on the x-axis at a distance $\bar{x}_{\text{surf}} = \frac{\pi}{2}$ from the y-axis, i.e. at $\left(\frac{\pi}{2}, 0\right)$.

2. The arc length of the curve can be calculated using the formula

$$\begin{aligned} K &= \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx \\ &= \int_0^1 \sqrt{1+25}\,dx \\ &= \int_0^1 \sqrt{26}\,dx \\ &= \sqrt{26} \end{aligned}$$